

Sprint 4 Documentation – OCR worker, MinIO

Team: Sebastian and Felix

Repository: <https://github.com/BunnySupreme/SWEN3Project>

Sprint duration: 20.11.2025 – 27.11.2025

Estimated workload: 12 h / person / week

1. Sprint Goal (from assignment)

1. Create the dedicated OCR Worker application running as a separate service.
2. Integrate an OCR engine (Tesseract/Ghostscript or equivalent) and demonstrate it with unit tests.
3. Extend the REST server to store uploaded PDF documents in MinIO (S3-compatible storage).
4. Implement the OCR Worker to:
 - a. consume OCR job messages from RabbitMQ,
 - b. fetch the original PDF from MinIO,
 - c. perform OCR and generate a summary,
 - d. publish the OCR result back to RabbitMQ / REST server.
5. Extend `docker-compose.yml` to run MinIO and the OCR Worker in containers.

2. Implemented Work

OCR Worker (OcrWorkerService)

1. Implemented as a hosted service in a dedicated Worker project.
2. Consumes messages from paperless.ocr.input via RabbitMQ.
3. Uses IMinioClient/IFileStoreClient to download the original PDF from MinIO.
4. Uses IOcrEngine/Tesseract wrapper to perform OCR on the downloaded file.
5. Builds a ResultModel with DocumentId, Summary, and ProcessedAt.
6. Publishes the result as JSON to paperless.ocr.results.
7. Implements ACK/NACK and logging via log4net.

MinIO Integration

1. Added paperless-minio service to docker-compose.yml with persistent volume.
2. Exposed S3 API on port 9000 and console on port 9001.
3. Added environment variables for MinIO endpoint, credentials, and bucket name.
4. Adjusted REST API upload path to store file bytes in MinIO.

REST Server

1. On document upload:
 - a. Saves metadata to PostgreSQL.
 - b. Stores the PDF in MinIO.
 - c. Publishes an OCR job to RabbitMQ with `DocumentId` and object key.
2. RabbitConsumerService consumes OCR result messages and updates DocumentModel.Summary.

Testing

1. Added unit tests for:
 - OCR Worker's processing logic (given a message, calls MinIO and OCR engine, publishes result).
 - MinIO file-storage integration (mocking S3 client).

6. Workload (Estimate)

Team Member	Week / Sprint	Estimated Hours	Main Tasks
Sebastian	Sprint 4	12	docker-compose
Felix	Sprint 4	12	